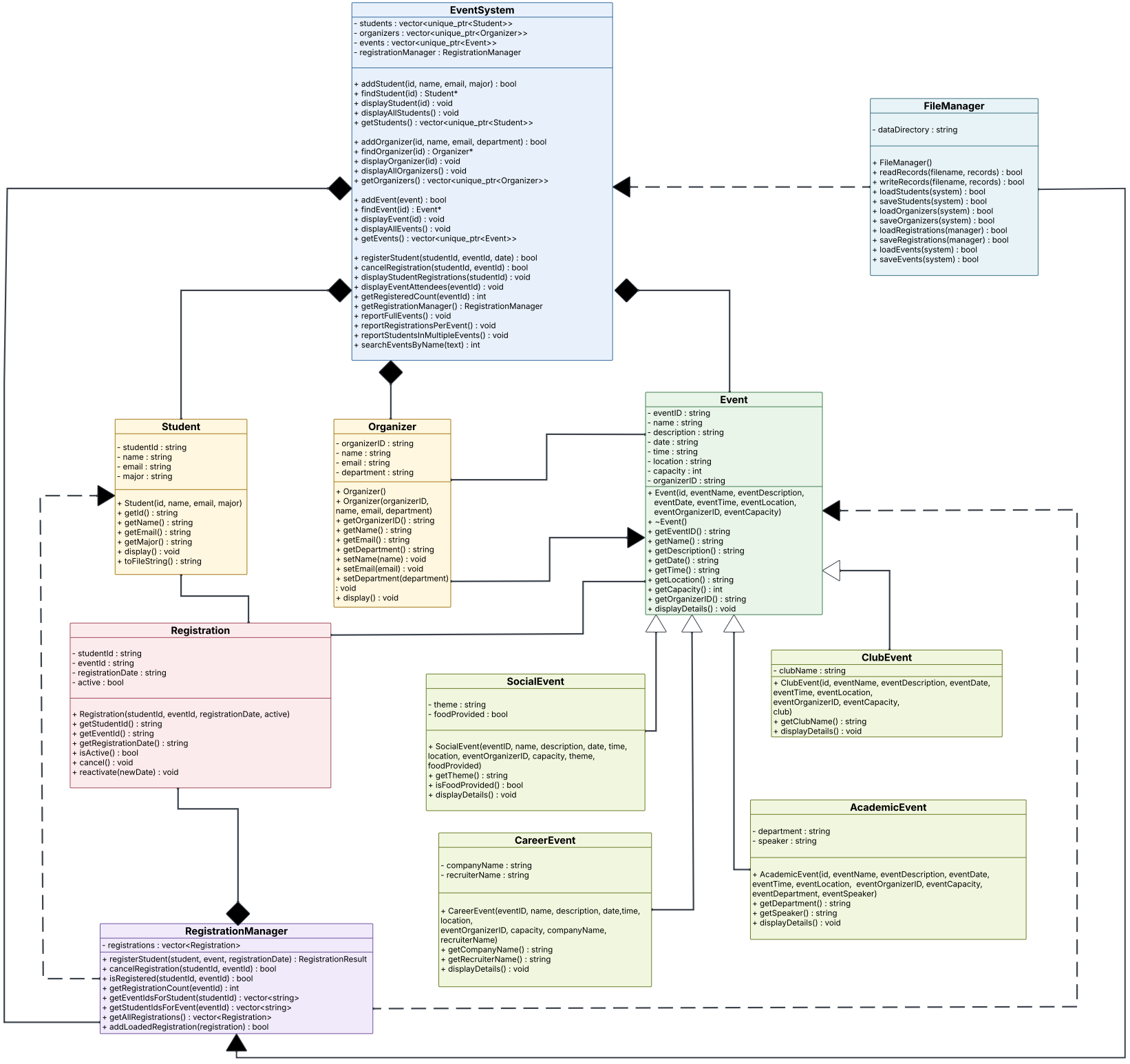

EventSystem
- students : vector<unique_ptr<Student>>
- organizers : vector<unique_ptr<Organizer>>
- events : vector<unique_ptr<Event>>
- registrationManager : RegistrationManager
+ addStudent(id, name, email, major) : bool
+ findStudent(id) : Student*
+ displayStudent(id) : void
+ displayAllStudents() : void
+ getStudents() : vector<unique_ptr<Student>>
+ addOrganizer(id, name, email, department) : bool
+ findOrganizer(id) : Organizer*
+ displayOrganizer(id) : void
+ displayAllOrganizers() : void
+ getOrganizers() : vector<unique_ptr<Organizer>>
+ addEvent(event) : bool
+ findEvent(id) : Event*
+ displayEvent(id) : void
+ displayAllEvents() : void
+ getEvents() : vector<unique_ptr<Event>>
+ registerStudent(studentId, eventId, date) : bool
+ cancelRegistration(studentId, eventId) : bool
+ displayStudentRegistrations(studentId) : void
+ displayEventAttendees(eventId) : void
+ getRegisteredCount(eventId) : int
+ getRegistrationManager() : RegistrationManager
+ reportFullEvents() : void
+ reportRegistrationsPerEvent() : void
+ reportStudentsInMultipleEvents() : void
+ searchEventsByName(text) : int
FileManager
- dataDirectory : string
+ FileManager()
+ readRecords(filename, records) : bool
+ writeRecords(filename, records) : bool
+ loadStudents(system) : bool
+ saveStudents(system) : bool
+ loadOrganizers(system) : bool
+ saveOrganizers(system) : bool
+ loadRegistrations(manager) : bool
+ saveRegistrations(manager) : bool
+ loadEvents(system) : bool
+ saveEvents(system) : bool
Student
- studentId : string
- name : string
- email : string
- major : string
+ Student(id, name, email, major)
+ getId() : string
+ getName() : string
+ getEmail() : string
+ getMajor() : string
+ display() : void
+ toFileString() : string
Organizer
- organizerID : string
- name : string
- email : string
- department : string
+ Organizer()
+ Organizer(organizerID, name, email, department)
+ getOrganizerID() : string
+ getName() : string
+ getEmail() : string
+ getDepartment() : string
+ setName(name) : void
+ setEmail(email) : void
+ setDepartment(department) : void
+ display() : void
Event
- eventID : string
- name : string
- description : string
- date : string
- time : string
- location : string
- capacity : int
- organizerID : string
+ Event(id, eventName, eventDescription, eventDate, eventTime, eventLocation, eventOrganizerID, eventCapacity)
+ ~Event()
+ getEventID() : string
+ getName() : string
+ getDescription() : string
+ getDate() : string
+ getTime() : string
+ getLocation() : string
+ getCapacity() : int
+ getOrganizerID() : string
+ displayDetails() : void
Registration
- studentId : string
- eventId : string
- registrationDate : string
- active : bool
+ Registration(studentId, eventId, registrationDate, active)
+ getStudentId() : string
+ getEventId() : string
+ getRegistrationDate() : string
+ isActive() : bool
+ cancel() : void
+ reactivate(newDate) : void
SocialEvent
- theme : string
- foodProvided : bool
+ SocialEvent(eventID, name, description, date, time, location, eventOrganizerID, capacity, theme, foodProvided)
+ getTheme() : string
+ isFoodProvided() : bool
+ displayDetails() : void
ClubEvent
- clubName : string
+ ClubEvent(id, eventName, eventDescription, eventDate, eventTime, eventLocation, eventOrganizerID, eventCapacity, club)
+ getClubName() : string
+ displayDetails() : void
CareerEvent
- companyName : string
- recruiterName : string
+ CareerEvent(eventID, name, description, date,time, location, eventOrganizerID, capacity, companyName, recruiterName)
+ getCompanyName() : string
+ getRecruiterName() : string
+ displayDetails() : void
AcademicEvent
- department : string
- speaker : string
+ AcademicEvent(id, eventName, eventDescription, eventDate, eventTime, eventLocation, eventOrganizerID, eventCapacity, eventDepartment, eventSpeaker)
+ getDepartment() : string
+ getSpeaker() : string
+ displayDetails() : void
RegistrationManager
- registrations : vector<Registration>
+ registerStudent(student, event, registrationDate) : RegistrationResult
+ cancelRegistration(studentId, eventId) : bool
+ isRegistered(studentId, eventId) : bool
+ getRegistrationCount(eventId) : int
+ getEventIdsForStudent(studentId) : vector<string>
+ getStudentIdsForEvent(eventId) : vector<string>
+ getAllRegistrations() : vector<Registration>
+ addLoadedRegistration(registration) : bool